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Roll No

EX-5004-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Power Electronic Devices and Circuits

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What is IGBT? What are its other names? Give its basic structural features and working. 7
b) Compare power MOSFET with BJT. Also discuss transfer and output characteristics of power MOSFET. 7
2. a) Describe turn-on methods of thyristor. Also draw its V-I characteristics. 7
b) For an SCR, the gate-cathode characteristics has a straight line slope of 130. For trigger source voltage of 15V and allowable gate power dissipation of 0.5 watts, compute the gate-source resistance. 7
3. a) Describe the over voltage protection of thyristors in detail. 7
b) Explain class F commutation (Line Commutation) for commutating thyristor. 7

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4. a) Explain the principle and operation of single phase half wave thyristor rectifier circuit with waveforms. 7
b) Give the comparison of mid point and bridge rectifier circuit? 7
5. a) Describe the working of single phase full bridge inverter. 7
b) Explain the means for reduction of harmonics in the inverter output voltage. 7
6. Explain the operation of three phase bridge inverter for 180° mode along with necessary data and waveforms. 14
7. a) Describe the principle of operation of step down chopper. 7
b) A step up chopper has input voltage of 220V and output voltage of 660 V. If the non-conducting time of thyristor-chopper is 100 μ s, compute the pulse width of output voltage.
In case pulse width is halved for constant frequency operation, Find the new output voltage. 7
8. Explain the following. (any two) 14
 - a) Single phase voltage controller with R load
 - b) Buck and Boost converter
 - c) Dual converter
